# STAFF REVIEW · H446 085 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

## Do Now — source answers

1. The input must be sorted.

2. O(n log n).

3. f=g+h.

4. Pop removes C then B; the final stack is [A,D] bottom-to-top.

## I Do — source worked steps

1. Method and result — Binary search O(log n) is valid because identifiers are sorted; if indexed dictionary is available, keyed lookup is another option.

## We Do — guided CHECK

1. Method and result
Dijkstra finds least-cost path with non-negative edge weights; do not confuse with array search.

## You Do — source mark schemes

1. Linear [2], binary [2], Dijkstra [2] with correct contexts.

2. Bubble $O(n^2)$, O(1) [2]; insertion $O(n^2)$,O(1) [2]; merge O(n log n),O(n) [2]; quick $O(n^2)$ worst, recursion stack depends on partitions [2].

3. One-off linear or indexed lookup on original unsorted input [2], sorting/indexing requires preprocessing [1], repeated sorted searches permit binary O(log n) [2], data changes may demand index maintenance [1].

## Exit Ticket — source answer

1. False. Preconditions and data structure matter.